Reg. No. | | | | | | | | | | | [18MTEEM201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE - 2019

DESIGN OF WATER AND WASTE WATER TREATMENT SYSTEMS

Time: 03 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full Questions. Question No. 1 and 2 are compulsory.**
ii. Missing data may be suitably assumed.

1.	**a.**	What chemicals cause hardness in water and how is hardness determined in laboratory testing?	**12**
	b.	Define Biochemical Oxygen Demand. With standard notations derive an expression for BOD.	**08**
2.	**a.**	Explain the following physical water treatment principles: **i.** Reverse osmosis **ii.** Membrane separation	**10**
	b.	Explain any two types of water reactors in brief.	**10**
3.	**a.**	Explain the following principles of chemical treatment: **i.** Co-agulation **ii.** Flocculation	**10**
	b.	Explain the design of municipal waste water treatment plant.	**10**
		OR	
4.	**a.**	How chemical coagulation works. Explain in brief.	**10**
	b.	Surface water is coagulated by adding 30 mg/l of ferrous sulfate and an equivalent dosage of lime. How many pounds of coagulant are used per million gallons of water processed? How many pounds of lime are required at a purity of 80 percent CaO?	**10**
5.	**a.**	Draw the flow chart and explain residue management system.	**08**
	b.	Explain with sketches the working principle and construction of industrial waste water system.	**12**
		OR	
6.	**a.**	What are the key elements of security to be considered in protecting waterworks?	**10**
	b.	Explain the concept of decision tree for ultimate disposal of bio-solids.	**10**
7.	**a.**	With respect to design of waste water treatment plant briefly explain the following: **i.** Grid chamber **ii.** Settling tank sludge **iii.** Sludge dewatering system **iv.** Sludge drain bed	**20**
		OR	
8.	**a.**	Write a note on recent trends in design of municipal waste water treatment plant.	**10**
	b.	Discuss a case study related to waste water treatment plant system.	**10**

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